Module 2: SRS & PRD Document

Driver Dashboard & Live GPS Tracking Architecture

Module Name: Mobile Driver Telemetry & Roster
Target Platform: Flutter Mobile (Android/iOS)
Document Type: Combined SRS & PRD
Version: v2.1 Final

1. EXECUTIVE SUMMARY & DRIVER OVERVIEW

The Driver Dashboard operates on an in-bus smartphone/tablet. It manages morning route initiation, broadcasts live GPS telemetry adhering to the 100-meter movement rule, renders 10 assigned stop sequences, and regulates passenger boarding via a 2-minute waiting timer.

2. ASSIGNED BUS & MORNING JOURNEY EXECUTION

- **Fixed Assignment Lock:** Drivers view pre-assigned Bus Registration (e.g., KA 02 AB 1234) and Route ID (e.g., Route 5). Drivers cannot edit vehicle or route parameters.
- **Journey Lifecycle:** Driver presses **[START MORNING JOURNEY]**, triggering a modal confirmation and initiating background GPS tracking.
- **Profile & RC View:** Driver can open overlays to review profile metadata and Vehicle RC registration documents stored in Firebase Storage.

3. THE 100-METER DISTANCE MOVEMENT RULE LOGIC

To prevent database clutter and smooth out GPS jitter, location updates are published only when distance delta $\ge 100\text{ meters}$.

```
[ GPS Sensor Captures Coordinates (lat_2, lon_2) ] | [ Compute Haversine Distance (d) from Last Publis
```

4. PASSENGER BOARDING & TWO-MINUTE WAITING ALGORITHM

At each of the 10 stops, passengers are grouped into three status categories:

Status Category	Indicator Color	Driver Boarding Rule
Coming	GREEN	Confirmed boarding. Driver waits for passenger to arrive and board.
Absent	RED	Explicitly marked absent. Driver immediately skips waiting logic.
Not Updated	AMBER	No status submitted. System executes a mandatory **2-minute countdown timer** upon stop arrival.

5. TWO-MINUTE TIMER EXECUTION WORKFLOW

```
[ Driver Confirms Arrival at Stop N ] | Are there "Not Updated" Passengers? / \ [ YES ] [ NO ] | | [ S
```

6. LIVE MAP & DELAY CALCULATION LOGIC

- **Road-Following Polyline:** Maps connect 10 stops along actual roads using the Google Maps Directions API rather than straight lines.
- **Dynamic Marker Display:** Renders a Bus Emoji icon when speed $> 3\text{ km/h}$, and a circle node when stationary.
- **Delay Telemetry:** Computed as $\text{Delay} = \text{Estimated Arrival Time} - \text{Scheduled Time}$. If delay $> 3\text{ mins}$, a delay icon and reason banner are displayed.

7. FIRESTORE DATABASE STRUCTURE FOR DRIVER TELEMETRY

Collection Path	Document Keys	Description
`liveBusLocations/{busId}`	`latitude`, `longitude`, `speed`, `journeyStatus`, `updatedAt`	Real-time location stream published under the 100m rule.
`buses/{busId}`	`registrationNumber`, `routeNumber`, `driverId`, `rcDocumentUrl`	Vehicle assignment metadata.
`passengerCheckins/{checkinId}`	`passengerId`, `busId`, `stopId`, `status` (`coming`/`absent`)	Shared check-in state linked with Student Dashboard.